

A Pre-Hospital Early Warning System for Ischemic Stroke Risk Based on Multi-Wavelength PPG and an XGBoost Algorithm

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I. INTRODUCTION

Ischemic stroke is one of the leading causes of long-term neurological disability and death worldwide. According to the Global Stroke Fact Sheet 2025, approximately 7.25 million deaths occur each year due to stroke [1]. Of these total cases, ischemic strokes account for approximately 65% globally and can reach as high as 87% in certain populations, such as in the United States [1], [2]. This clinical condition is highly time-sensitive; consequently, the greatest obstacle to effective treatment (based on the "time is brain" principle) often occurs during the prehospital phase. This delay is generally caused by

families' lack of awareness in recognizing the urgency of intermittent symptoms such as a Transient Ischemic Attack (TIA) [3], [4]. Although clinical risk stratification methods such as the ABCD2 score have been proven valid for evaluating the level of urgency following a TIA [5], these tools are generally only applied by medical personnel after the patient arrives at a healthcare facility. Therefore, there is a need for a continuous pre-hospital monitoring system capable of detecting early risk indicators and triggering automatic alerts, thereby accelerating the family's decision to seek immediate care at a healthcare facility.

Wearable devices that incorporate photoplethysmography (PPG) technology open up opportunities for non-invasive monitoring of physiological parameters [6]. Compared to single-optical sensors, signal acquisition using a multi-wavelength PPG platform takes advantage of the varying light penetration depths in skin tissue, resulting in more comprehensive and stable hemodynamic monitoring [7]. However, when predicting stroke risk based on health data, machine learning models generally face the challenge of extreme class imbalance in medical datasets [8], [9]. A literature review demonstrates that implementing a class-balanced loss function and data balancing techniques in the eXtreme Gradient Boosting (XGBoost) algorithm can significantly boost model performance in detecting the minority class (at-risk patients) [10], [11], [12], [13], [14].

Based on this literature review, the objective of this study is to design a distributed early-warning system architecture that integrates multi-wavelength PPG signal acquisition with

artificial intelligence based on the XGBoost algorithm. This classification model is specifically optimized using class weighting parameters to detect the risk of ischemic stroke in highly imbalanced data. The primary benefit expected from this study is the development of a wearable screening device with high sensitivity, which is integrated

II. METHOD

This study proposes a distributed early-warning system architecture for detecting the risk of ischemic stroke in the prehospital phase. The methodology applied in the development of this system is divided into three main stages: designing the hardware architecture for multi-wavelength PPG signal acquisition, the signal preprocessing stage along with the estimation of physiological parameters, and risk level classification modeling using the XGBoost algorithm.

A. Multi-Wavelength Signal Acquisition Architecture

The early warning system prototype is designed to continuously acquire the user's hemodynamic data using photoplethysmography (PPG) technology. To overcome the diagnostic limitations of a single sensor, the system integrates two optical sensors with different wavelengths (multi-wavelength). The SON1303 sensor is used on the green channel (525 nm) to detect changes in superficial capillary blood volume, while the MAX30102 sensor is used on the red (660 nm) and infrared (880 nm) channels to achieve deeper tissue penetration [6], [7]. The combination of these multiple light spectra aims to produce a more informative signal that is resistant to motion artifacts during monitoring. The entire process including data acquisition, three-channel PPG data sampling, and batch data organization, is controlled locally using an ESP32-S3 microcontroller.

B. Signal Preprocessing and Physiological Estimation

To minimize the computational load on the microcontroller (edge device), the system employs a distributed processing architecture between the prototype and the server. The ESP32-S3 is responsible for performing an initial signal quality assessment using the Signal Quality Index (SQI). Batches of optical data that meet the quality standards are then sent to the server for further preprocessing. The signals are filtered using a Butterworth bandpass filter (0.5–5 Hz) to isolate the pulsatile AC components from high-frequency noise, followed by Fast Fourier Transform (FFT) autocorrelation to calculate basic metrics.

The cleaned optical signals are then used to estimate five physiological parameters (systolic blood pressure, diastolic blood pressure, blood glucose, cholesterol, and uric acid). The mapping of the optical signal to these clinical parameters is performed using a combination of five separate regression models: Support Vector Regression (SVR) and eXtreme Gradient Boosting (XGBoost). To ensure that the trained regression models do not experience data leakage between users, model performance is rigorously evaluated using the Leave-One-Subject-Out (LOSO) cross-validation method. The results of these physiological parameter estimates are then

integrated with user profiles to generate nine risk factors that serve as inputs for the final stroke risk classification model.

C. Risk Level Classification Modeling Using XGBoost Algorithm

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III. RESULT AND DISCUSSION

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ACKNOWLEDGMENT

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IV. CONCLUSION

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